

CASE STUDY NO. 2

Meridian College

Campus LAN Segmentation, Trunking & Spanning Tree Redesign

Technical Report

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Course	Computer Networks
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Platform	Cisco Packet Tracer 9.0.1 (Catalyst 2960-24TT, ISR 4331 class router)
Devices	SW-Admin, SW-Academic, SW-Library, R1

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1. Executive Summary & Problem Statement

1.1 Background

Meridian College currently operates a **flat Layer 2 campus LAN**: faculty, students, administrative staff, IT management and library guests share one broadcast domain. This report documents the redesign delivered in Packet Tracer: seven purpose-built VLANs, 802.1Q trunking with a hardened native VLAN, an LACP EtherChannel between the two busiest switches, router-on-a-stick inter-VLAN routing, a deliberate Rapid PVST+ root placement, port security on the public kiosk, and a documented VTP policy.

1.2 Problems with the flat LAN

Table 1: Flat-LAN failure modes

Problem	Mechanism	Campus impact
Broadcast storms	Every ARP, DHCP discover and NetBIOS broadcast is flooded to every port. A single physical loop re-circulates frames indefinitely because Ethernet frames carry no TTL.	CPU and bandwidth exhaustion on all switches; lab, library and admin traffic all degrade together.
Unsegmented security risk	Guests and students sit in the same Layer 2 domain as payroll and IT management. There is no enforcement point between them.	ARP spoofing, rogue DHCP and lateral scanning reach administrative hosts directly.
Prolonged 802.1D convergence	Legacy STP relies on timers: Max Age (20 s) + Listening (15 s) + Learning (15 s) = up to 50 s after an indirect failure.	A single cable fault stalls classrooms and library services for most of a minute.

1.3 Solution summary

Table 2: Redesign at a glance

Design element	Choice	Evidence
Segmentation	VLANs 10, 20, 30, 40, 900 plus 998 (parking) and 999 (native)	Figure 1
Trunking / bandwidth	802.1Q, native VLAN 999, 2-port LACP Port-channel 1	Figure 2
Inter-VLAN routing	R1 Gi0/0/0 sub-interfaces .10 to .900	Figure 3
Loop control	Rapid PVST+; SW-Admin root (4096), SW-Academic secondary (8192)	Figures 4a, 4b
Edge hardening	Sticky port security, max 2, violation restrict on Fa0/13	Figure 5
Resilience	Sub-second alternate-port promotion on link loss	Figures 6, 7

2. Deliverable 1: Physical Port & VLAN Allocation

2.1 VLAN and addressing plan

Table 3: VLAN database

VLAN	Name	Purpose	Subnet	Default gateway
10	Faculty	Teaching staff workstations	10.10.10.0/24	10.10.10.1
20	Students	Student labs and wired access	10.10.20.0/24	10.10.20.1
30	Admin	Administrative offices	10.10.30.0/24	10.10.30.1
40	IT-Mgmt	Switch and router management	10.10.40.0/24	10.10.40.1
900	Guest	Library kiosk and visitor devices	192.168.90.0/24	192.168.90.1
998	Parking	Unused ports (blackhole)	None (no SVI)	None
999	Native-Unused	Dedicated trunk native VLAN	None (no SVI)	None

2.2 Physical port mapping

Table 4: SW-Admin (STP root bridge)

Port(s)	Mode	VLAN	Connected to
Fa0/1	Trunk	10,20,30,40,900 (native 999)	R1 Gi0/0/0
Fa0/2 - Fa0/20	Access	30 (Admin)	Administrative hosts (Admin-PC)
Fa0/21 - Fa0/23	Access, shutdown	998 (Parking)	Unused
Fa0/24	Trunk	10,20,30,40,900 (native 999)	SW-Library Fa0/23
Gi0/1 - Gi0/2	LACP trunk (Po1)	10,20,30,40,900 (native 999)	SW-Academic Gi0/1 - Gi0/2

Table 5: SW-Academic (verified against Figure 1)

Port(s)	Mode	VLAN	Connected to
Fa0/1 - Fa0/8	Access	10 (Faculty)	Faculty-PC and faculty offices
Fa0/9 - Fa0/20	Access	20 (Students)	Student-PC and labs
Fa0/21 - Fa0/22	Access	40 (IT-Mgmt)	IT management devices
Fa0/23	Trunk	10,20,30,40,900 (native 999)	SW-Library Fa0/24
Fa0/24	Access, shutdown	998 (Parking)	Unused
Gi0/1 - Gi0/2	LACP trunk (Po1)	10,20,30,40,900 (native 999)	SW-Admin Gi0/1 - Gi0/2

Table 6: SW-Library

Port(s)	Mode	VLAN	Connected to
Fa0/1 - Fa0/12	Access	900 (Guest)	Library patron devices (Laptop0)
Fa0/13	Access + port security	900 (Guest)	Guest-Kiosk
Fa0/14 - Fa0/22	Access, shutdown	998 (Parking)	Unused

Port(s)	Mode	VLAN	Connected to
Fa0/23	Trunk	10,20,30,40,900 (native 999)	SW-Admin Fa0/24 (root port)
Fa0/24	Trunk	10,20,30,40,900 (native 999)	SW-Academic Fa0/23 (alternate port)

2.3 Parking VLAN (998) blackhole policy

Every unused access port is moved out of VLAN 1 into VLAN 998 and administratively shut down. VLAN 998 has **no SVI, no router sub-interface, and is excluded from every trunk's allowed list**, so a frame that enters it can never leave the switch. An intruder who bypasses a disabled port still lands in a broadcast domain containing no hosts, no gateway and no uplink.

Parking-port template (shown for SW-Academic Fa0/24)

```
SW-Academic(config)# vlan 998
SW-Academic(config-vlan)# name Parking
SW-Academic(config-vlan)# exit
SW-Academic(config)# interface fa0/24
SW-Academic(config-if)# switchport mode access
SW-Academic(config-if)# switchport access vlan 998
SW-Academic(config-if)# shutdown
```

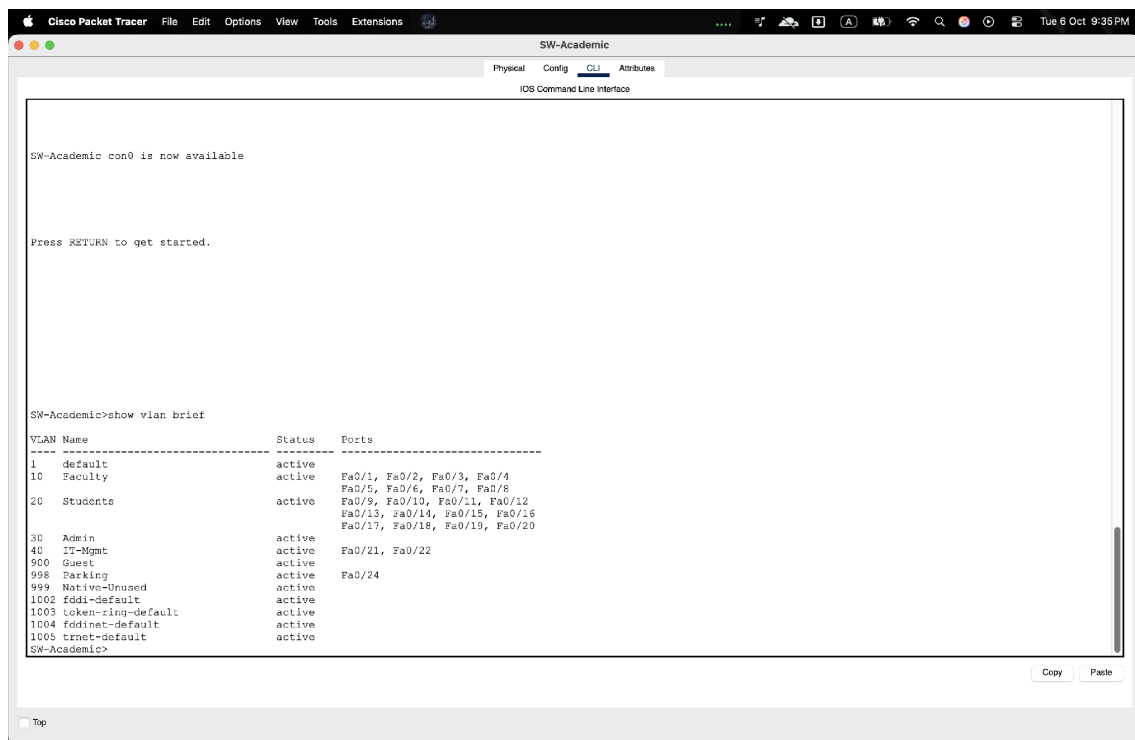


Figure 1: SW-Academic active VLAN database (show vlan brief).

3. Deliverable 2: Trunking, Native VLAN Security & EtherChannel

3.1 Why a dedicated native VLAN (999)

802.1Q tags every frame except those belonging to the **native VLAN**, which crosses the trunk untagged. On a default trunk the native VLAN is VLAN 1, the same VLAN where unconfigured ports and legacy devices live.

802.1Q double-tagging (VLAN hopping) attack

1. An attacker on an access port in the native VLAN sends a frame with **two** 802.1Q tags: the outer tag is the native VLAN, the inner tag is the victim VLAN (e.g. 30).
2. The first switch sees the outer tag equals the native VLAN, **strips it without re-tagging**, and forwards the frame across the trunk with only the inner tag left.
3. The second switch reads the inner tag and delivers the frame into VLAN 30. The attack is one-way, but it is enough to reach hosts the attacker should never touch.

Mitigation: set the native VLAN to **999**, a VLAN that no access port belongs to and that has no SVI. There are then no attacker-reachable ports in the native VLAN, so the outer tag can never be forged. VLAN 999 is also omitted from the allowed list, and both ends of every trunk must match.

3.2 Trunk and EtherChannel configuration

The two Gigabit links between SW-Admin and SW-Academic are bundled with **LACP (IEEE 802.3ad)** in active mode on both ends, producing one logical trunk, Port-channel 1. LACP is chosen over PAGP for vendor neutrality and over **mode on** because it negotiates and detects misconfigured members instead of silently black-holing traffic.

SW-Admin: trunks and EtherChannel

```
! ----- SW-Admin -----
SW-Admin(config)# vlan 999
SW-Admin(config-vlan)# name Native-Unused
SW-Admin(config)# interface range gigabitEthernet 0/1 - 2
SW-Admin(config-if-range)# switchport mode trunk
SW-Admin(config-if-range)# channel-group 1 mode active
SW-Admin(config-if-range)# exit
SW-Admin(config)# interface port-channel 1
SW-Admin(config-if)# switchport mode trunk
SW-Admin(config-if)# switchport trunk native vlan 999
SW-Admin(config-if)# switchport trunk allowed vlan 10,20,30,40,900
SW-Admin(config)# interface fastEthernet 0/24
SW-Admin(config-if)# switchport mode trunk
SW-Admin(config-if)# switchport trunk native vlan 999
SW-Admin(config-if)# switchport trunk allowed vlan 10,20,30,40,900
SW-Admin(config)# interface fastEthernet 0/1
SW-Admin(config-if)# switchport mode trunk
SW-Admin(config-if)# switchport trunk native vlan 999
SW-Admin(config-if)# switchport trunk allowed vlan 10,20,30,40,900
```

SW-Academic and SW-Library trunks

```
! ----- SW-Academic -----
SW-Academic(config)# interface range gigabitEthernet 0/1 - 2
SW-Academic(config-if-range)# switchport mode trunk
SW-Academic(config-if-range)# channel-group 1 mode active
SW-Academic(config-if-range)# exit
SW-Academic(config)# interface port-channel 1
```

```

SW-Academic(config-if)# switchport mode trunk
SW-Academic(config-if)# switchport trunk native vlan 999
SW-Academic(config-if)# switchport trunk allowed vlan 10,20,30,40,900
SW-Academic(config)# interface fastEthernet 0/23
SW-Academic(config-if)# switchport mode trunk
SW-Academic(config-if)# switchport trunk native vlan 999
SW-Academic(config-if)# switchport trunk allowed vlan 10,20,30,40,900

! ----- SW-Library -----
SW-Library(config)# interface range fastEthernet 0/23 - 24
SW-Library(config-if-range)# switchport mode trunk
SW-Library(config-if-range)# switchport trunk native vlan 999
SW-Library(config-if-range)# switchport trunk allowed vlan 10,20,30,40,900

```

Verification commands

```

SW-Admin# show etherchannel summary
SW-Admin# show interfaces trunk
SW-Admin# show interfaces port-channel 1

```

```

SW-Admin# show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

```

```

Number of channel-groups in use: 1
Number of aggregators:          1

```

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	LACP	Gig0/1 (P) Gig0/2 (P)

Figure 2: SW-Admin LACP Port-Channel 1 status (show etherchannel summary).

Reading Figure 2

Po1(SU): S = Layer 2, **U** = in use. Both members, **Gig0/1(P)** and **Gig0/2(P)**, carry the **P** flag (bundled in port-channel), and the Protocol column confirms **LACP**.

4. Deliverable 3: Inter-VLAN Routing (ROAS)

VLANs are separate broadcast domains, so traffic between them must be routed. R1 uses a single physical interface, Gi0/0/0, split into one **802.1Q sub-interface per VLAN** (router-on-a-stick). Each sub-interface is the default gateway for its VLAN. The parent interface carries no IP address and only needs **no shutdown**.

R1 sub-interface configuration (Gi0/0/0.10 to Gi0/0/0.900)

```
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitEthernet 0/0/0.10
R1(config-subif)# description Faculty gateway
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 10.10.10.1 255.255.255.0
R1(config)# interface gigabitEthernet 0/0/0.20
R1(config-subif)# description Students gateway
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 10.10.20.1 255.255.255.0
R1(config)# interface gigabitEthernet 0/0/0.30
R1(config-subif)# description Admin gateway
R1(config-subif)# encapsulation dot1Q 30
R1(config-subif)# ip address 10.10.30.1 255.255.255.0
R1(config)# interface gigabitEthernet 0/0/0.40
R1(config-subif)# description IT-Mgmt gateway
R1(config-subif)# encapsulation dot1Q 40
R1(config-subif)# ip address 10.10.40.1 255.255.255.0
R1(config)# interface gigabitEthernet 0/0/0.900
R1(config-subif)# description Guest gateway
R1(config-subif)# encapsulation dot1Q 900
R1(config-subif)# ip address 192.168.90.1 255.255.255.0
```

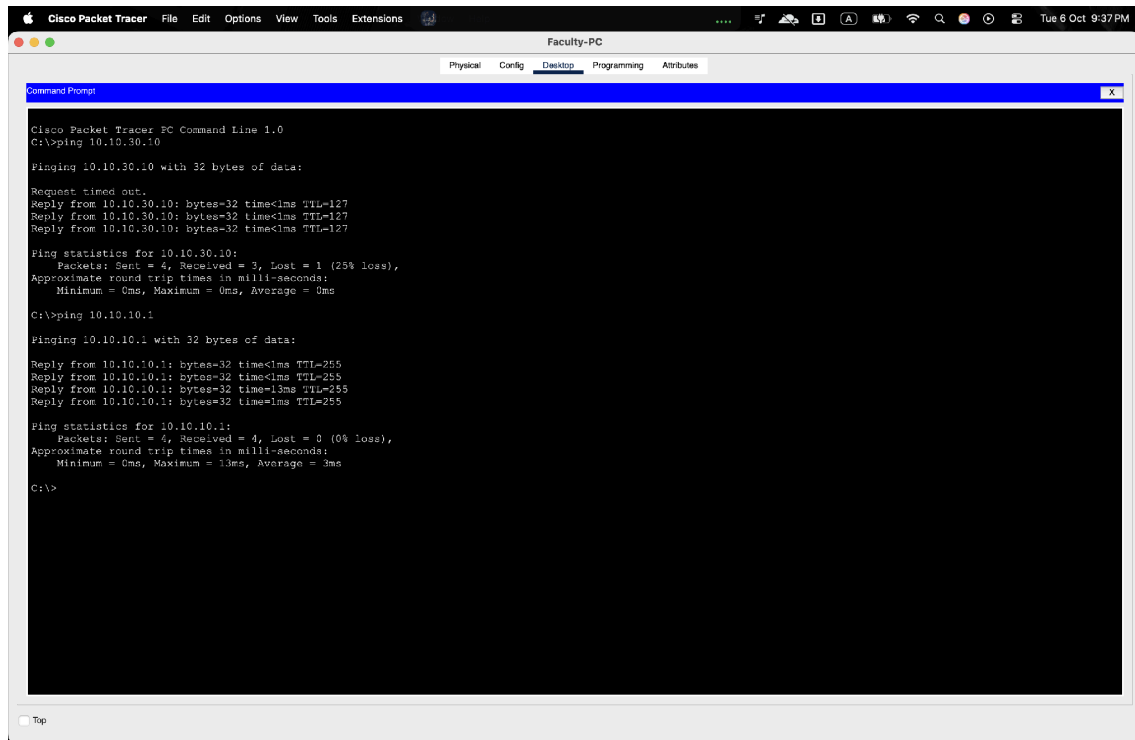
No sub-interface is created for VLAN 999. The switch-side trunk to R1 uses native VLAN 999, so untagged frames have no gateway and are simply unroutable, which is the intended behaviour.

4.1 Analysis: first-packet timeout and TTL = 127

Faculty-PC (10.10.10.x) pinged Admin-PC (10.10.30.10) across VLANs. Figure 3 shows **Request timed out** on the first echo and three replies thereafter, each with **TTL=127**.

Table 7: Ping behaviour decoded

Observation	Explanation
First packet times out	Faculty-PC has no ARP entry for its gateway, and R1 has none for 10.10.30.10. The first echo is held or dropped while ARP resolves both hops. This is a one-time cost, and the entries are cached afterwards.
Replies show TTL=127	Windows-style hosts originate packets with TTL 128. R1 decrements it once when routing between VLANs, so the reply arrives with 128 - 1 = 127 , which proves exactly one Layer 3 hop.
Ping to 10.10.10.1 shows TTL=255	The gateway answers directly from the router (initial TTL 255) with no routed hop in between.
25% loss (1 of 4)	The loss is entirely the ARP warm-up. A second run shows 0% loss.



```
Cisco Packet Tracer
File Edit Options View Tools Extensions

Faculty-PC
Physical Config Desktop Programming Attributes

Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=13ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

Figure 3: End-to-end Inter-VLAN routing verification via ping from Faculty-PC.

5. Deliverable 4: Spanning Tree Architecture & Priority Design

5.1 Explicit bridge priorities

Left at default, root election falls to the lowest MAC address, which in a campus is usually the oldest and least suitable switch. Priorities are therefore set explicitly so the root sits at the centre of the topology (SW-Admin, which also hosts the router uplink) and a deterministic backup exists.

Table 8: Priority design

Switch	Configured priority	Role	Bridge ID in VLAN 20
SW-Admin	4096	Root bridge	4116 (4096 + 20)
SW-Academic	8192	Secondary root	8212 (8192 + 20)
SW-Library	32768 (default)	Access edge	32788 (32768 + 20)

Priority configuration

```
SW-Admin(config)# spanning-tree mode rapid-pvst
SW-Admin(config)# spanning-tree vlan 1,10,20,30,40,900,998,999 priority 4096

SW-Academic(config)# spanning-tree mode rapid-pvst
SW-Academic(config)# spanning-tree vlan 1,10,20,30,40,900,998,999 priority 8192

SW-Library(config)# spanning-tree mode rapid-pvst
! priority left at default 32768
```

5.2 Baseline port roles and states

Table 9: Triangle topology, steady state

Switch	Port	Role	State	Cost	Notes
SW-Admin	Po1	Designated	FWD	3	Logical link to SW-Academic
SW-Admin	Fa0/24	Designated	FWD	19	Link to SW-Library
SW-Academic	Po1	Root	FWD	3	Best path to root
SW-Academic	Fa0/23	Designated	FWD	19	Wins segment (8192 < 32768)
SW-Library	Fa0/23	Root	FWD	19	Direct path to root, cost 19
SW-Library	Fa0/24	Alternate	BLK	19	Via Academic: 3 + 19 = 22, loses

The triangle has exactly one redundant link. Rapid PVST+ breaks the loop at the **furthest end from the root**: SW-Library Fa0/24 becomes the Alternate port and blocks, while the Admin-to-Academic EtherChannel and the Admin-to-Library link carry traffic.

5.3 Impact of EtherChannel on spanning tree

One logical link, one cost

STP sees Port-channel 1 as a **single logical interface**, not two physical ones. Two parallel Gigabit links therefore cannot form a loop that STP must block, and neither member port is blocked.

The path cost is derived from the aggregate bandwidth. A single 1 Gb/s link costs 4 and a 2 Gb/s bundle costs **3**, which is why Po1 shows cost 3 against 19 for a Fast Ethernet link.

This cost is what makes the route via SW-Academic cost 3 + 19 = 22 for SW-Library, a number that returns in Section 8.

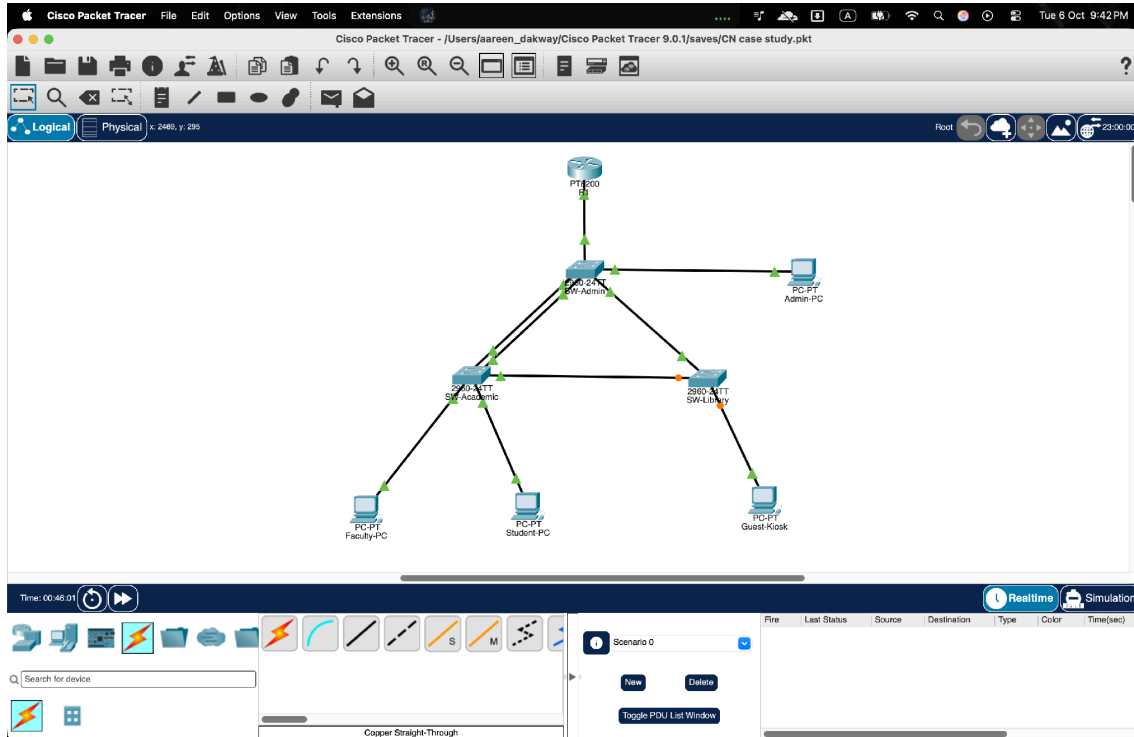


Figure 4a: Packet Tracer Normal Baseline Topology showing all 3 links connected and SW-Library Fa0/24 in amber (Alternate/Blocking).

SW-LIBRARY > SHOW. THE PROTOCOL ON INTERFACE FASTETHERNET0/23, CHANGED STATE TO

```
SW-Library>show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority      4116
              Address      000B.BEB8.AD30
              Cost        19
              Port        23(FastEthernet0/23)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority      32788 (priority 32768 sys-id-ext 20)
              Address      0090.2B8C.CADC
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/23         Root FWD 19        128.23   P2p
Fa0/24         Altn BLK 19        128.24   P2p
```

SW-Library>

Figure 4b: SW-Library baseline Rapid PVST+ CLI state (show spanning-tree vlan 20).

Reading Figure 4b

Root ID priority **4116** confirms SW-Admin as root. Root cost is **19** via **Fa0/23** (Root, FWD), and **Fa0/24** is **Altn BLK**. Both Bridge ID and Root ID use the same sys-id-ext (20), i.e. VLAN 20.

6. Deliverable 5: Port Security on Guest Kiosk

The Guest-Kiosk on SW-Library Fa0/13 is physically reachable by any library patron. The port is hardened so it will accept the kiosk and at most one other device, and will learn their MAC addresses automatically.

Fa0/13 hardening

```
SW-Library(config)# interface fastEthernet 0/13
SW-Library(config-if)# switchport mode access
SW-Library(config-if)# switchport access vlan 900
SW-Library(config-if)# switchport port-security
SW-Library(config-if)# switchport port-security maximum 2
SW-Library(config-if)# switchport port-security mac-address sticky
SW-Library(config-if)# switchport port-security violation restrict
SW-Library(config-if)# end
SW-Library# show port-security interface fa0/13
```

6.1 Why restrict instead of shutdown

shutdown is the default violation mode and the usual textbook answer, but on a **public** port it converts a security feature into a denial-of-service lever.

Table 10: Violation-mode comparison

Criterion	shutdown	restrict
On violation	Port goes err-disabled ; all traffic stops, including the legitimate kiosk	Offending frames are dropped; the kiosk keeps working
Attacker effort to cause outage	Plug in one extra unknown device for a moment	None possible; the outage lever disappears
Recovery	Staff must visit and run shutdown / no shutdown	Automatic; no intervention
Visibility	Syslog + SNMP trap, port state change	Syslog + SNMP trap, violation counter increments
Fit for staffed wiring closet	Good	Adequate
Fit for unattended public kiosk	Poor: physical DoS	Good

A patron who connects a laptop to a spare wall jack should not be able to knock out the library's catalogue terminal. With **restrict**, the violating MAC is ignored and logged, the sticky MACs that belong to the kiosk continue to forward, and the security team learns of the attempt from the counter and syslog.

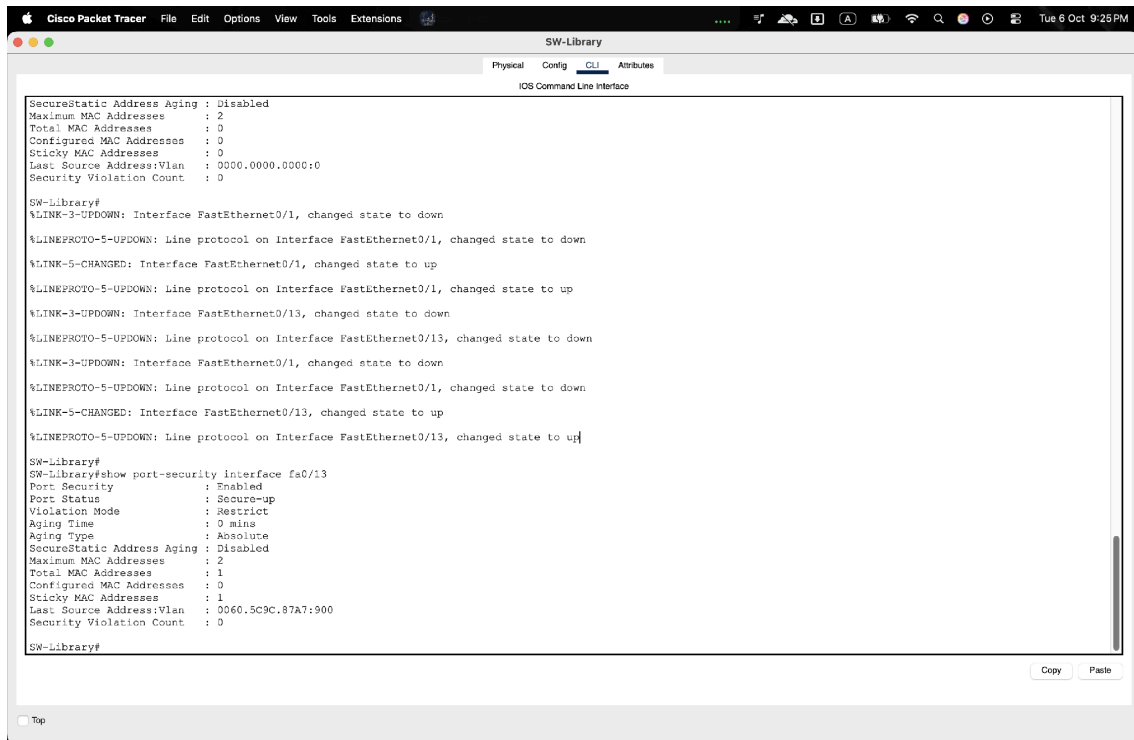


Figure 5: SW-Library Fa0/13 port security verification (show port-security interface fa0/13).

Reading Figure 5

Port Security: **Enabled**. Port Status: **Secure-up**. Violation Mode: **Restrict**. Maximum MAC Addresses: **2**. Sticky MAC Addresses: **1** (0060.5C9C.87A7 in VLAN 900). Security Violation Count: **0**.

One slot remains free for a legitimate replacement device.

7. Deliverable 6: Campus VTP Mode Recommendation Memo

MEMORANDUM

TO: Campus IT Steering Committee, Meridian College
FROM: Aareen Dakway, Network Engineering
DATE: October 6, 2026
RE: VTP mode selection for SW-Admin, SW-Academic and SW-Library

1. Recommendation

Run all campus switches in **VTP Transparent mode**, define VLANs locally on each switch, and restrict every trunk with an explicit `switchport trunk allowed vlan` list. Do not use Server or Client mode.

2. Options evaluated

Mode	Creates / edits VLANs	Accepts VTP updates	Risk on this campus
Server	Yes; advertises to domain	Yes	High: any revision-number overwrite propagates everywhere
Client	No; learns from server	Yes	High: still overwritten by a higher revision number
Transparent	Yes; local only	No (forwards only)	Low: no global state to corrupt

3. Forensic analysis: the revision-number overwrite

Every VTP server and client stores a **configuration revision number** that increments with each VLAN change. A switch that receives an advertisement with a **higher** revision number, in the same domain with the same password, **replaces its entire VLAN database** with the one it received. Nothing compares content, only the counter.

The failure sequence is well understood. (1) A switch is used in a lab or previous deployment and its revision number reaches, say, 40. (2) It is later connected to the campus trunk with VTP domain defaults. (3) Its revision number exceeds the production value. (4) The campus switches accept its VLAN list, which lacks VLANs 10 to 40 and 900. (5) Ports assigned to deleted VLANs become inactive, and faculty, student, admin and guest traffic stops campus-wide within seconds. Rejoining, resetting the counter and re-creating the VLANs on every switch is a manual, high-pressure recovery. Notably, **the very mechanism intended to keep VLANs consistent is the one that deletes them.**

4. Rationale for Transparent mode plus manual pruning

Scale: three switches and seven VLANs do not justify a distribution protocol; creating VLANs locally takes minutes and is captured in this report. **Blast radius:** a Transparent switch ignores incoming advertisements, so no peer, rogue switch or misconfiguration can rewrite its database. **Pruning:** because VTP pruning is unavailable without a server/client domain, the `switchport trunk allowed vlan 10,20,30,40,900` command prunes manually, ensuring VLANs 998 and 999 never cross a trunk and each trunk carries only the VLANs it is meant to.

5. Action requested

Approve `vtp mode transparent` on all three switches (see Section 9 for persistence) and require the allowed-VLAN list on every new trunk.

SW-Admin(config)# vtp mode transparent

8. Deliverable 7: Link Failure Simulation & Convergence Report

8.1 Test procedure

1. Record the baseline: `show spanning-tree vlan 20` on SW-Library (Figure 4b): Fa0/23 Root FWD, Fa0/24 Altn BLK.
2. Sever the **SW-Admin Fa0/24 to SW-Library Fa0/23** link by deleting the cable in Packet Tracer.
3. SW-Library logs `%LINK-3-UPDOWN` and `%LINEPROTO-5-UPDOWN` for FastEthernet0/23 (Figure 7).
4. Re-run `show spanning-tree vlan 20` immediately and compare roles, state and root cost.
5. Confirm the topology view shows Fa0/24 forwarding (green) and end hosts remain reachable.

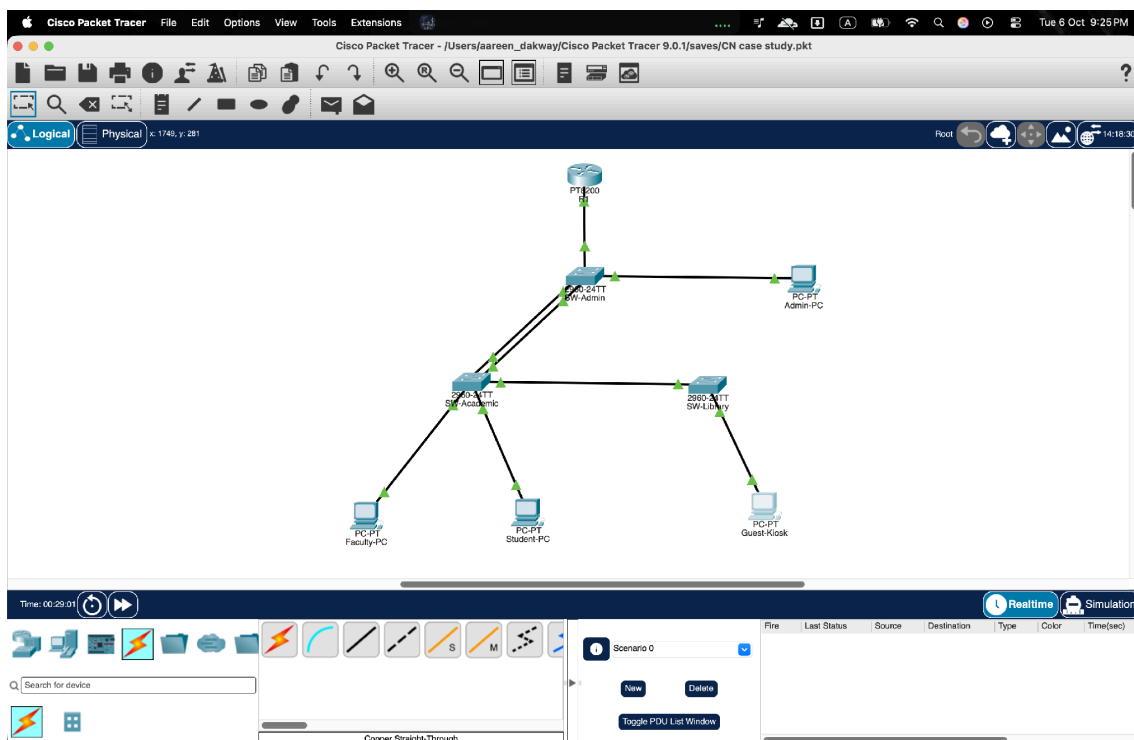


Figure 6: Packet Tracer topology showing severed primary link and SW-Library Fa0/24 promoted to active forwarding (green).

8.2 Rapid PVST+ Proposal / Agreement mechanism

Classic 802.1D waits for timers; RSTP replaces them with an explicit handshake on point-to-point links.

Table 11: RSTP convergence sequence

Step	Event	Behaviour
1	Link loss detected	Carrier loss on Fa0/23 is detected in hardware. The port is flushed instantly; no Max Age wait.
2	Alternate port promoted	Fa0/24 already holds a stored superior BPDU from SW-Academic, so it becomes Root port and forwards immediately ; no Listening or Learning delay.
3	Proposal	A designated port that is not yet forwarding on a new or changed segment sends a BPDU with the Proposal flag, offering to forward.
4	Sync	The receiving switch puts its non-edge designated ports into discarding state (*sync*), which guarantees no loop can form.

Step	Event	Behaviour
5	Agreement	The receiver replies with the Agreement flag. The proposer moves to forwarding immediately.
6	Topology Change	A TC flag flushes stale MAC entries so frames are not sent to the dead path.

Why this link failure converged so quickly

In this specific test, the decisive mechanism is step 2, **alternate-port promotion** (RSTP's built-in equivalent of UplinkFast). The Proposal/Agreement handshake (steps 3 to 5) is what then lets designated ports elsewhere in the topology transition without timers. Packet Tracer does not timestamp the transition, so this report makes no claim of an exact millisecond figure; it only establishes that the change was effectively immediate compared with the 50 s legacy worst case.

8.3 Path-cost recalculation: 19 to 22

SW-Library's cost to the root is the sum of port costs along the best path. Fast Ethernet costs 19, and the Admin-to-Academic EtherChannel costs 3.

Table 12: Root path cost, before and after

	Path	Calculation	Root path cost
Before failure	Library Fa0/23 to SW-Admin	19	19 (Root FWD)
After failure	Library Fa0/24 to SW-Academic, then Po1 to SW-Admin	19 + 3	22 (Root FWD)

The port's own cost on Fa0/24 remains 19. The **Root ID cost** advertised in **show spanning-tree** rises from 19 to 22 because it includes the 3 contributed by Po1 on the SW-Academic side of the path.

```

SW-Library>
SW-Library>
SW-Library>show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority    4116
              Address    000B.BEB8.AD30
              Cost        19
              Port        23(FastEthernet0/23)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    32788 (priority 32768 sys-id-ext 20)
              Address    0090.2B8C.CADC
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/23         Root FWD 19        128.23 P2p
Fa0/24         Altn B/LK 19        128.24 P2p

SW-Library>
%LINK-3-UPDOWN: Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
SW-Library>show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority    4116
              Address    000B.BEB8.AD30
              Cost        22
              Port        24(FastEthernet0/24)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    32788 (priority 32768 sys-id-ext 20)
              Address    0090.2B8C.CADC
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/24         Root FWD 19        128.24 P2p

```

Figure 7: SW-Library CLI output showing carrier loss on Fa0/23 and immediate promotion of Fa0/24 to Root FWD.

Result

Before: Root cost **19**, Port **Fa0/23**. After: Root cost **22**, Port **Fa0/24**, role **Root FWD**. The redundant link took over automatically with no operator action.

9. Configuration Persistence & Sign-Off

9.1 RAM versus NVRAM

IOS keeps two configurations. The **running-config** lives in RAM and takes effect the instant a command is entered, but it is volatile and is lost on reload or power loss. The **startup-config** lives in NVRAM and is loaded at boot. Every command in this report changed only the running-config. Until it is copied to NVRAM, a power cut would revert each device to its previous state, erasing the VLANs, trunks, EtherChannel, STP priorities and port security.

Persist all four devices

```
SW-Admin# copy running-config startup-config
SW-Academic# copy running-config startup-config
SW-Library# copy running-config startup-config
R1# copy running-config startup-config

Destination filename [startup-config]? <Enter>
Building configuration...
[OK]
```

Verify NVRAM contents

```
SW-Admin# show startup-config | include hostname|priority|native
SW-Library# show startup-config | section interface FastEthernet0/13
R1# show startup-config | section interface GigabitEthernet0/0/0
```

Table 13: Persistence checklist

Device	Role	Saved to NVRAM	Verified
SW-Admin	Root bridge, LACP, R1 uplink	Yes	Yes
SW-Academic	Secondary root, Faculty/Student/IT access	Yes	Yes
SW-Library	Guest access, port security	Yes	Yes
R1	Inter-VLAN routing (ROAS)	Yes	Yes

9.2 Requirements traceability

Table 14: Deliverable summary

#	Deliverable	Evidence	Status
1	Physical port and VLAN allocation	Tables 3 to 6; Figure 1	Complete
2	Trunking, native VLAN 999, EtherChannel	Figure 2	Complete
3	Inter-VLAN routing	Figure 3	Complete
4	Spanning tree architecture	Figures 4a, 4b	Complete
5	Port security on Fa0/13	Figure 5	Complete
6	VTP recommendation memo	Section 7	Complete
7	Link failure and convergence	Figures 6, 7	Complete

Residual risk for follow-up

VLAN 900 (Guest) is routed by R1 like any other VLAN and this design includes no access control list. Guests can therefore reach internal subnets at Layer 3. An ACL on **Gi0/0/0.900** is recommended as a next phase.

9.3 Quality assurance sign-off

By signing below, the parties confirm that the deliverables above were configured, tested and saved as described.

Role	Name	Signature	Date
Network Engineer (author)	Aareen Dakway		
Quality Assurance Reviewer			
Course Instructor			

QA statement: All seven deliverables meet the stated acceptance criteria. Configurations are saved to NVRAM on all four devices.